MAIS PAC

IPhO 04.04: Approximation, Taylor Series and Numerical Methods · Worked Solutions

Approximation, Taylor Series and Numerical Methods: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Small-angle estimates. Return →

(a) 3.0° = 0.05236 {rad}, so sin 3.0° ≈ 0.05236.

(b) The exact value is 0.05234; the error is 2×10^(-5), i.e. 0.004%.

(c) The relative error is about θ²/6, so 1% requires θ = √(0.06) = 0.245 {rad} = 14°. **Check:** the cubic term θ³/6 at θ = 0.0524 is 2.4×10^(-5), matching the error found in (b) — the series is doing exactly what it promises.

Solution 2 — The binomial approximation. Return →

(a) g = (GM)/((R+h)²) = g_0(1+(h)/(R))^(-2).

(b) For h≪ R, (1+(h)/(R))^(-2) ≈ 1 - (2h)/(R), so (Δ g)/(g_0) ≈ -(2h)/(R).

(c) With h = 400 {km} and R = 6400 {km}, 2h/R = 0.125, so g falls by about 12.5%, to 8.6 {m,s^(-2)}. **Check:** the next term is 3(h/R)² = 0.012, an extra 1%, so the first-order approximation is good to about 1% — worth quoting.

Solution 3 — Which term dominates? Return →

(a) sin θ ≈ θ - (θ³)/(6).

(b) The θ³ term is a perturbation of relative size θ_0²/6 acting on an oscillator, and its effect on the period is of order θ_0² — the standard result is T = T_0(1+θ_0²/16), so the coefficient is 1/16, not 1/6, but the order is set by θ_0².

(c) θ_0²/16 = 0.01 gives θ_0 = 0.40 {rad} = 23°. **Check:** this is why pendulum clocks were built with small amplitudes and why the 1% correction only appears above about 20°.

Solution 4 — Newton–Raphson by hand. Return →

(a) Solve f(x) = x²-10 = 0, so x_(n+1) = x_n - (x_n²-10)/(2x_n) = (1)/(2)(x_n+(10)/(x_n)).

(b) x_0 = 3: x_1 = (1)/(2)(3+3.333) = 3.1667; x_2 = (1)/(2)(3.1667 + 3.1579) = 3.1623.

(c) The error satisfies e_(n+1)≈ (f'')/(2f')e_n², i.e. it is squared each step; since e≈10^(-2) becomes 10^(-4), the number of correct digits doubles. **Check:** √(10) = 3.16228, and two iterations from a starting guess of 3 already give five correct digits.

Solution 5 — Bisection. Return →

(a) f(1) = -1 and f(2) = 5; the function is continuous, so by the intermediate value theorem a root lies between them.

(b) f(1.5) = 0.875>0, so the root is in (1, 1.5). f(1.25) = -0.297<0, so it is in (1.25, 1.5). f(1.375) = 0.225>0, so it is in (1.25, 1.375). The root is 1.325.

(c) Each step halves the interval, so n steps give an interval of 1/2^n; six decimal places needs 1/2^n < 10^(-6), i.e. n ≈ 20. Bisection is used when reliability matters more than speed — it cannot diverge and needs no derivative. **Check:** 1.325³-1.325-1 = -0.002, so the bracket is correct.

Solution 6 — Round-off and catastrophic cancellation. Return →

(a) √(1.0001) = 1.0000499988, and subtracting 1 leaves 0.0000499988 — only a few of the original seven digits survive, because the leading digits cancel.

(b) (1+ε)^(1/2) ≈ 1 + ε/2 with ε = 10^(-4), so the difference is 5.000×10^(-5), with full accuracy.

(c) The rule is to rewrite the expression so that no subtraction of nearly equal quantities occurs — rationalise, expand, or factor out the small term before computing. **Check:** the analytic result 5.0×10^(-5) matches the naive computation to three digits, but the naive route could have been far worse.

Solution 7 — A Fermi estimate. Return →

(a) Air density at atmospheric pressure is about 1.2 {kg,m^(-3)} (or use pV = nRT with p≈10^5 {Pa} and M≈29 {g,mol^(-1)}).

(b) Volume = 5×4×3 = 60 {m³}, so mass = 1.2×60 ≈ 70 {kg}.

(c) The density is the uncertain number, since it depends on temperature and pressure; a 10% error in density gives a 10% error in mass, while the room dimensions are good to a few percent. **Check:** 70 {kg} of air in a small room is about one person's mass — a memorable, and correct, order of magnitude.

Solution 8 — A small correction with a sign. Return →

(a) In the accelerating frame the effective gravity is g_({eff}) = g+a.

(b) T = 2π√(L/(g+a)) = T_0(1+(a)/(g))^(-1/2) ≈ T_0(1 - (a)/(2g)), so the fractional change is -a/2g.

(c) At a = 0.10g the approximation gives -5.0%, whereas the exact value is 1/√(1.1)-1 = -4.65%; the difference is 0.35%, the size of the neglected (3/8)(a/g)² term. The sign is negative because a larger effective gravity shortens the period, as a stiffer spring would. **Check:** the second-order term is 3a²/8g² = 0.00375, matching the discrepancy.

Solution 9 — Iterating a transcendental equation. Return →

(a) The current is the same through the diode and resistor: I_s(e^(V/V_T)-1) = (1.5-V)/(1000).

(b) The unknown appears both in an exponential and linearly, so no algebraic rearrangement isolates it.

(c) **Guess the diode is off:** then V = 1.5 {V} would drive I = I_se^(57.7) ≈ 10^(-12)×10^(25) {A}, absurdly large — so the diode is strongly on. **Assume the diode drops about 0.7 {V}:** then I ≈ (1.5-0.7)/1000 = 0.8 {mA}, and check V = V_Tln (I/I_s) = 0.026ln (8×10^8) = 0.54 {V} — inconsistent, so the drop is nearer 0.54 {V}. Iterating once more with I = (1.5-0.54)/1000 = 0.96 {mA} gives V = 0.55 {V}, converged. The diode is in forward conduction. **Check:** the two estimates 0.54 and 0.55 {V} differ by less than the diode's own temperature variation, so the operating point is (0.55 {V}, 0.95 {mA}).

Solution 10 — Choosing a method. Return →

(a) Let f(x) = x-cos x; f(0) = -1<0 and f(1) = 0.46>0, so a root lies in (0,1).

(b) With g(x) = cos x, |g'(x)| = |sin x| sin 1 = 0.84<1 on the interval, so the iteration converges. Starting from x_0 = 0.7: 0.765, 0.722, 0.750, 0.732, 0.743, 0.736 — converging on x = 0.739.

(c) The convergence factor is about 0.67 per step, so each iteration gains about 0.17 of a decimal digit; Newton–Raphson would gain digits quadratically. In an exam, three or four fixed-point iterations with a table is faster to write and quite sufficient; Newton–Raphson is preferable if high precision is demanded. **Check:** cos 0.739 = 0.7391, which confirms the fixed point.